

Asymmetric impacts of foreign direct investment, economic growth, fossil fuel dependence on renewable energy consumption in Thailand

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ABSTRACT

Thailand's transition toward a sustainable energy system has become a central concern in both academic discourse and national policymaking, particularly as the country faces rising energy demand, dependence on imported fossil fuels, and international commitments to emissions reduction. Despite increasing attention to renewable energy (RE) and foreign direct investment (FDI), empirical evidence on how these factors interact with economic growth (EG) remains fragmented. This study addresses this gap by combining econometric analysis with energy system modeling to examine the dynamic relationships among EG, FDI, fossil fuel (FFC) use, and RE consumption in Thailand from 1990 to 2023. Using a VECM framework, the historical analysis reveals that RE consumption has not yet exerted a statistically significant influence on EG, while FDI and FFC remain dominant drivers. To complement these historical findings and assess future transitions, the study integrates LEAP and MARKAL models to simulate technology-specific energy pathways and system-wide optimization scenarios to 2050. The results show that targeted green FDI, combined with efficiency-enhancing energy system design, can substantially increase renewable penetration and reduce emissions, even though the historical contribution of renewables has been modest. The study provides evidence-based policy insights for aligning investment strategies, energy planning, and decarbonization targets, and it highlights the need for more advanced modeling approaches to capture nonlinear and structural dynamics in Thailand's evolving energy landscape. The findings suggest that Thai policymakers should prioritize green FDI incentives, RE infrastructure investment, and grid modernization policies to accelerate the transition to a more efficient, low-carbon energy system while sustaining long-term EG.

1. Introduction

Thailand's energy-economic landscape provides strong empirical justification for examining the interactions among EG, FDI, FFC, RE development, and CO₂ emissions. Over the past three decades, Thailand has experienced steady economic expansion, with GDP increasing more than fourfold since the early 1990s, accompanied by a substantial rise in energy demand across all sectors. The country remains heavily dependent on imported FFCs—particularly natural gas and oil—which account for more than two-thirds of total primary energy supply, making economic performance highly sensitive to global energy price fluctuations. At the same time, Thailand has positioned FDI as a central

component of its development strategy, with foreign investment increasingly directed toward energy infrastructure, manufacturing, and high-technology industries. However, RE has grown more slowly, accounting for a modest share of the national energy mix for most of the study period, despite recent policy initiatives to expand solar, biomass, and wind capacity. These dynamics underscore the empirical relevance of jointly analyzing FDI, FFC use, RE consumption, and EG to better understand Thailand's current development trajectory and its challenges in meeting long-term sustainability and decarbonization goals [1–4].

Thailand has experienced rapid EG growth over the past few decades, leading to increased energy demand, primarily from FFCs. Despite this dependency, Thailand has set ambitious targets for REC

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expansion to reduce its carbon footprint while ensuring energy security. The country's REC sector, particularly solar and biomass, has witnessed significant growth in recent years, yet FFCs remain dominant in its energy mix. Balancing EG with energy transition presents a complex challenge that requires careful policy planning and investment strategies. FDI is another critical factor influencing Thailand's energy sector. FDI, particularly in energy-intensive industries such as manufacturing, has contributed to the increased consumption of FFCs. At the same time, FDI can catalyze REC development by bringing in capital, technology, and expertise. The interplay between FDI and EC-FFC-based and RE is critical in understanding Thailand's economic and environmental trajectory [5,6].

The manuscript explores the interrelation among EG, FDI, FFC, and REC in Thailand. The study analyzes how these variables influence Thailand's economic and environmental future. Specifically, the research addresses the following questions: What is the impact of FDI on Thailand's Energy use patterns, particularly in terms of FFC and REC use? How does energy use from FFCs and renewable sources affect EG in Thailand? Can Thailand transition towards REC without compromising its EG? The manuscript examines the short-term and long-term relationships among these variables through a detailed econometric analysis, offering insights into how Thailand can achieve sustainable economic development while transitioning to a low-carbon economy. The results will help policymakers design strategies to foster EG, attract FDI, and promote REC, thereby ensuring a more sustainable energy future for the country. The manuscript aims to investigate the interrelationship between EG, FDI, FFC, and REC consumption in Thailand [7,8].

This study contributes to the existing literature in several important ways. Unlike previous studies that primarily rely on conventional econometric approaches to examine the relationships among EG, FDI, fossil fuel consumption (FFC), and RE consumption, the present research integrates econometric analysis with energy system modeling to provide both historical and forward-looking insights into Thailand's energy transition. Specifically, the study combines ARDL-VECM estimation with LEAP and MARKAL simulations, enabling the analysis to move beyond identifying statistical relationships to evaluate future technology pathways, energy-system efficiency, and emissions outcomes under different policy and investment scenarios. In addition, the study employs an updated dataset covering the period 1990–2023, which captures recent structural changes in Thailand's energy sector, including the expansion of RE technologies and evolving FDI patterns. Furthermore, the research contributes to the policy literature by translating empirical and scenario-based findings into technology-specific and system-oriented policy recommendations related to green investment, renewable deployment, and long-term energy planning. By integrating macroeconomic analysis with engineering-based energy modeling, the study offers a more comprehensive framework for understanding sustainable energy transitions in emerging economies and therefore extends the current body of knowledge beyond traditional single-method analyses [9,10].

Despite substantial research on the relationships among EG, energy consumption (EC), and environmental quality, empirical evidence for Thailand remains fragmented and incomplete, particularly regarding the combined roles of FDI, FFC dependence, and RE development. Existing studies typically examine these variables in isolation, employ purely econometric approaches that do not link them to energy system behavior, or overlook the technological pathways through which structural changes in the energy sector might influence long-run economic outcomes. As a result, the literature has not yet provided a comprehensive assessment that integrates historical dynamics with forward-looking energy system simulations to inform Thailand's transition toward cleaner, more resilient energy systems. This study addresses this gap by combining econometric techniques (VECM) with engineering-based LEAP and MARKAL modeling to analyze both historical interactions and future technological pathways. The contribution

is twofold: academically, the work advances the literature by bridging economic and energy-system perspectives within a unified framework; and practically, it provides policymakers with evidence-based insights into how FDI allocation, energy diversification, and renewable technology deployment could shape Thailand's long-term sustainability and development strategy. This combined motivation—both theoretical and practical—demonstrates the need for the present study and its contribution to ongoing academic and policy debates [11,12].

The motivation for this study arises from Thailand's increasing energy demand, continued dependence on FFs, growing environmental pressures, and the need to achieve sustainable EG while meeting long-term decarbonization commitments. Although FDI and RE development are frequently promoted as important drivers of sustainable transition, the existing literature provides mixed and fragmented evidence regarding their actual contributions to EG and environmental sustainability in Thailand [13–15].

Moreover, previous studies have generally relied on conventional econometric approaches without integrating future-oriented energy-system modeling capable of evaluating long-term technological and policy transition pathways. In response to these gaps, the present study pursues the following objectives: (1) to examine the long-run and short-run relationships among EG, FDI, FF consumption, RE consumption, and CO₂ emissions in Thailand during the period 1990–2023; (2) to evaluate how RE and FF dependence influence Thailand's economic and environmental performance; (3) to integrate econometric analysis with LEAP and MARKAL energy-system modeling in order to simulate future energy transition scenarios; and (4) to provide evidence-based and technology-oriented policy recommendations that support sustainable energy planning, green investment, and long-term energy security in Thailand. These objectives provide a clearer analytical direction for the study and strengthen the connection between the research motivation, theoretical framework, methodology, and policy relevance [1,16].

2. Theoretical framework

Understanding the interrelationships among EG, FDI, FFC, REC, and environmental outcomes requires a clear theoretical basis. This section outlines the primary theoretical strands that justify the hypotheses examined in this study and clarifies the mechanisms through which these variables are expected to interact [17,18].

2.1. Energy-growth nexus theory

The traditional neoclassical and endogenous growth theories posit that energy is a critical input in the production process, alongside labor and capital. Under this framework, EC—particularly FFCs—enhances productive capacity by powering industrial, transportation, and manufacturing activities. The “growth hypothesis” states that higher EC drives economic expansion, while the “conservation hypothesis” suggests that growth influences energy demand. These conflicting views underscore the importance of empirically testing the direction and strength of the relationship between the variables [19,20].

For Thailand, where industrialization and urbanization have historically relied on FFC-based infrastructure, this theoretical foundation supports the expectation of a positive linkage between FFC and GDP growth [13,16].

2.2. The role of FDI in economic and energy transition theories

FDI is widely understood through the lens of the technology-transfer theory and the spillover effects framework, both of which argue that foreign investment enhances host-country productivity by introducing new technologies, managerial practices, and capital accumulation. In the energy sector, FDI can influence the economy through two channels. The traditional channel involves FDI inflows into energy-intensive manufacturing or FFC infrastructure, which may increase FFC

consumption and emissions, thereby supporting short-term growth but worsening environmental outcomes [14,15]. Green Channel: FDI directed toward RE technologies, efficiency improvements, and clean infrastructure can accelerate structural transformation by lowering energy system costs and modernizing the grid. According to endogenous growth theory, such technological advancements can generate long-run productivity gains. These theoretical mechanisms justify examining the dual role of FDI in shaping both economic expansion and energy transitions [21,22].

2.3. RE and sustainable growth theory

SD theory emphasizes the long-run benefits of transitioning from FFCs to RE sources. Modern theoretical perspectives argue that REC contributes to economic performance through energy diversification, reducing vulnerability to global oil price shocks; lower marginal costs of electricity, especially for solar and wind; innovation spillovers from clean-technology adoption; improved energy security; and reduced import dependence. However, the transition theory of energy systems also notes that renewable technologies require high upfront investment, infrastructure adaptation, and grid modernization—processes that may temporarily weaken or delay the observable impact of REC on GDP. This tension explains why the historical impact of REC on EG may be limited, even if its future potential is substantial [21,23,24].

2.4. Environmental Kuznets Curve (EKC) and emissions dynamics

The Environmental Kuznets Curve hypothesis provides a theoretical basis for the relationship between EG and CO₂ emissions. According to the EKC, emissions initially increase with income as economies industrialize, but eventually decline once higher-income stages enable cleaner technologies, stricter regulations, and energy-efficient practices. In Thailand's case—an upper-middle-income economy—emissions dynamics may still lie on the rising portion of the curve due to continued reliance on FFCs and incomplete integration of renewables. This theoretical insight suggests that EG may still contribute positively to emissions in the historical record [25–27].

2.5. Systems transition theory and integrated modeling rationale

Systems transition theory highlights how large-scale changes in energy systems emerge through interactions among technology, policy, market forces, and institutional factors. This theoretical perspective justifies integrating econometric models (which capture historical behavioral relationships) with energy system models like LEAP and MARKAL (which simulate future technological pathways under policy and investment scenarios) [27–29].

This combined theoretical logic supports the methodological approach, as econometric analysis identifies historical elasticities and directional relationships. LEAP/MARKAL simulations translate these relationships into future system-level outcomes. Together, these frameworks provide a theoretically grounded justification for analyzing how FDI, EC, and EG may jointly shape Thailand's long-run energy transition [22,30,31].

To provide a stronger theoretical foundation for the empirical analysis, the study's hypotheses are discussed separately and explicitly linked to the relevant theoretical perspectives. The expected positive relationship between FF consumption and EG is grounded in the energy–growth nexus theory, which argues that energy is a critical production input supporting industrialization and economic expansion [9,19,32]. The relationship between FDI and EG is explained by endogenous growth and technology-transfer theories, which posit that FDI contributes to productivity enhancement, capital accumulation, and technological spillovers. In contrast, the role of RE consumption is interpreted within the framework of sustainable transition theory, which emphasizes that renewable technologies may contribute to long-

run sustainability and energy security. However, their short-run macroeconomic effects may remain limited in economies with relatively low renewable penetration. Furthermore, the linkage between EG and environmental degradation is supported by the Environmental Kuznets Curve (EKC) hypothesis, which suggests that emissions tend to increase during early stages of industrial development before declining as cleaner technologies emerge. Based on these theoretical foundations, the conceptual framework presented in Fig. 1 visually illustrates the hypothesized interactions among EG, FDI, FF consumption, RE consumption, and CO₂ emissions, thereby providing a clearer analytical structure for the econometric and energy-system modeling employed in this study [10–12].

Based on the above theories, the following conceptual predictions guide the empirical analysis [33,34]:

H₁: FFC consumption positively contributes to EG (growth hypothesis).

H₂: RE has long-run growth potential, although historical effects may be weak (SD theory).

H₃: FDI affects growth either through energy-intensive growth or via clean-technology spillovers (technology-transfer theory).

H₄: EG may increase emissions depending on Thailand's EKC phase.

H₅: Integrated modeling is needed to translate historical relationships into future system design insights (systems transition theory).

The novelty of the present study is highlighted by several important distinctions from the existing literature on the energy–growth nexus. Previous studies on Thailand and other emerging economies have generally focused on isolated relationships, such as the impact of RE consumption on EG or the role of FF consumption in environmental degradation, using conventional econometric approaches alone. In contrast, this study develops a more comprehensive analytical framework by simultaneously examining the dynamic interactions among EG, FDI, FF consumption (FFC), RE consumption, and CO₂ emissions within a unified empirical and energy-system modeling structure [17,20,35]. Unlike earlier studies that rely exclusively on ARDL, VAR, or VECM methods, the present research integrates econometric analysis with LEAP and MARKAL simulations, thereby linking historical relationships to future energy-transition pathways and system-optimization scenarios [19,22,36,37].

Moreover, the study employs an updated dataset covering 1990–2023, enabling the analysis to capture Thailand's recent structural transformations, including RE expansion, green investment trends, and evolving decarbonization policies. This is particularly important given Thailand's ongoing transition under the Alternative Energy Development Plan (AEDP) and recent RE targets aimed at increasing RE shares and reducing FF dependence. Furthermore, while recent studies have increasingly examined the energy–environment nexus in European and advanced economies, relatively few studies have combined econometric evidence with engineering-based energy system modeling in the context of Southeast Asia and Thailand specifically. By integrating macroeconomic analysis with technology-oriented policy simulations, the present study provides a broader and more policy-relevant contribution to the literature on sustainable energy transitions in emerging economies [20,38,39].

The theoretical background of this study is grounded in several interrelated economic, energy, and environmental theories that explain the dynamic relationships among EG, FDI, FF consumption (FFC), RE consumption, and CO₂ emissions [40–42]. First, the energy–growth nexus theory suggests that energy consumption is a fundamental input for production and economic expansion, particularly in industrializing economies where FFs continue to dominate the energy structure. This theory supports the expectation that FF consumption contributes positively to EG. Second, endogenous growth theory and technology-transfer theory provide the conceptual basis for the relationship between FDI and economic performance, arguing that foreign investment promotes capital accumulation, technological innovation, productivity improvements, and infrastructure development. In the energy sector, FDI may

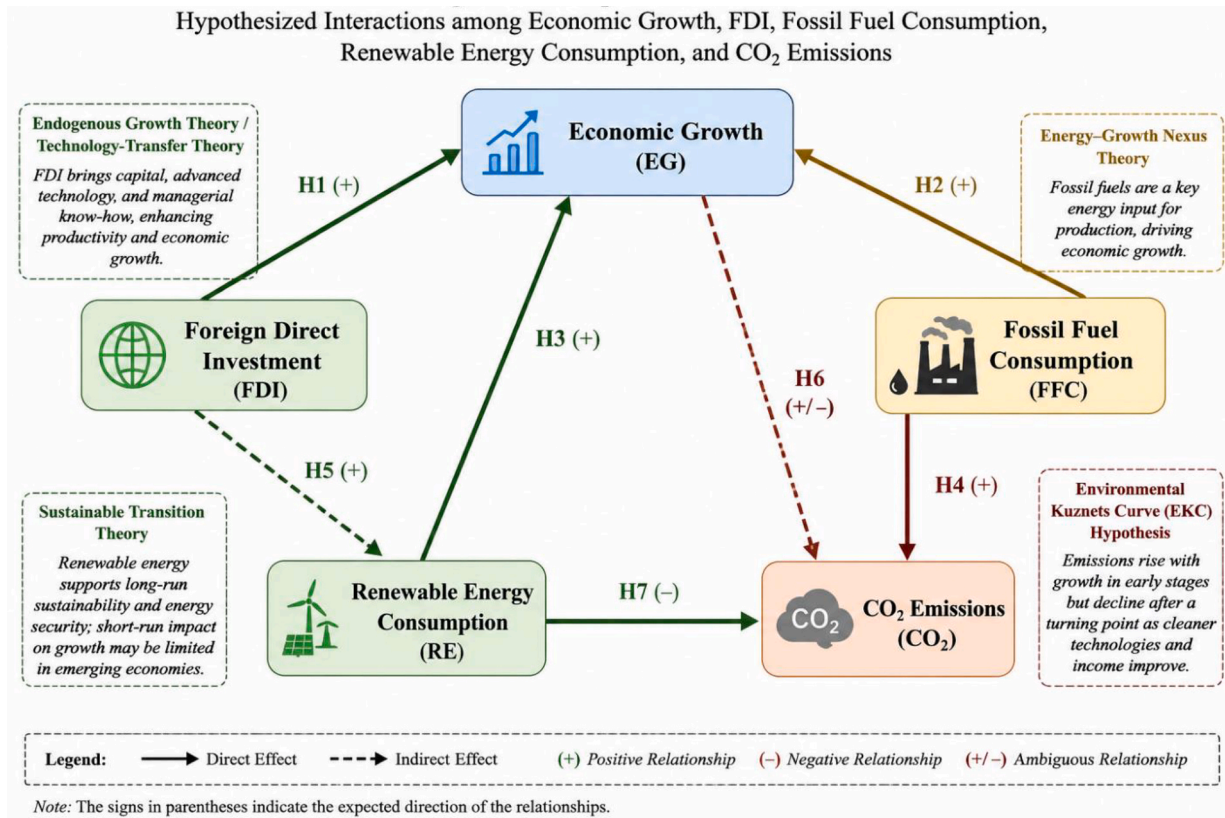


Fig. 1. Conceptual Framework.

also facilitate the diffusion of RE technologies and energy-efficient systems. Third, sustainable transition theory explains the growing importance of RE consumption in promoting long-term environmental sustainability, energy diversification, and reduced dependence on carbon-intensive fuels [38,39,43]. However, the theory also recognizes that RE may require time, technological maturity, and institutional support before generating significant macroeconomic impacts. Finally, the Environmental Kuznets Curve (EKC) hypothesis provides a theoretical explanation for the relationship between EG and environmental degradation, suggesting that emissions tend to rise during the early stages of industrial development before declining as economies adopt cleaner technologies and stronger environmental regulations. Together, these theoretical perspectives provide the conceptual foundation for the hypotheses and empirical analysis developed in this study [12,19].

3. Literature review

A substantial body of research examines the interrelationship between EG and EC. Early work showed a positive link between energy use (especially FFCs) and GDP growth. More recently, however, studies have sought to decouple EG from carbon-intensive energy sources. For example, in the EU-15, a 2025 panel ARDL analysis finds that RE consumption significantly reduces CO₂ emissions, even as EG remains carbon-intensive. Similarly, regional research indicates that growth alone no longer guarantees improved environmental performance, underscoring the need for technological and efficiency enhancements. These findings underscore the importance of investigating not only total EC but also its composition and efficiency in relation to growth [44–46].

The role of FDI in the energy–growth–investment nexus has garnered increasing attention, particularly under the lens of clean/green investment. Recent evidence from emerging markets indicates that a 1% increase in renewable-EC and FDI, combined, raises GDP by 1.02% and 1.87%, respectively, in the long run. [47] shows that FDI targeted at

green sectors accelerates the energy transition by facilitating capital inflows, technology transfer, and the uptake of infrastructure. These studies suggest that FDI is not only a growth driver but also an enabler of energy-system transformation, making it a critical variable in modeling transitions for countries like Thailand [48,49].

Previous studies have examined the interplay between EG, FDI, and EC. Studies by [50] suggest that FDI can significantly impact Energy use patterns, particularly in emerging economies where energy-intensive industries attract foreign investment. However, the specific role of REC in this context remains to be studied, especially in Thailand. Thailand's energy sector has historically relied on FFCs, with coal, oil, and natural gas being the predominant energy sources. Nevertheless, growing concerns over climate change and energy security have prompted Thailand to diversify its energy portfolio by promoting renewable energy sources such as solar, wind, and biomass. While evidence suggests that REC can contribute to EG [51], the extent to which it can replace FFCs in Thailand remains an open question—the interrelation between EG, EC, and FDI has been the subject of extensive research in recent decades. As countries aim for EG while addressing environmental concerns, the dynamics among these factors have become increasingly important. This section reviews relevant literature on the Interrelation between EG, FDI, FFC, and REC use, focusing on Thailand's unique context [52,53].

Numerous studies have explored the link between Energy use and EG. According to the energy-led growth hypothesis, energy is a critical input to economic production, and disruptions in energy supply may hinder economic activity [54]. Empirical studies in developing economies suggest that increased Energy use strongly correlates with GDP growth, particularly in energy-dependent sectors such as manufacturing and heavy industries [55]. For example, studies in countries such as China, India, and Malaysia have shown that EC—especially from FFCs—has historically driven rapid industrialization and urbanization [56]. Thailand's rapid EG has been energy-intensive, with FFCs

comprising most of its energy mix [57]. FFC in Thailand, particularly in the transportation and industrial sectors, has been identified as a critical driver of EG growth. However, Thailand's growing reliance on energy imports has raised concerns about energy security and the volatility of global fuel prices. In response, the Thai government has been promoting REC to reduce dependency on FFCs and diversify its energy sources [18,58,59].

FDI is vital to EG, especially in emerging markets like Thailand, and the interrelation between FDI and EC is, however, multifaceted. FDI inflows often lead to increased energy use as foreign investors invest in energy-intensive sectors, such as manufacturing, mining, and construction [60]. This influx of FDI can boost EG and lead to higher levels of FFC environmental degradation [61]. Studies by [62] suggest that FDI inflows can have positive and negative environmental consequences, depending on the type of investment. FDI is directed toward energy-efficient or REC projects that can reduce carbon footprints, while investments in traditional, energy-intensive industries can exacerbate environmental challenges. In Thailand, FDI has significantly contributed to EG. However, it has also increased FFC and attracted substantial FDI into energy-intensive sectors, thereby significantly affecting the country's energy use patterns. However, with the Thai government's push for REC, FDI has shifted towards clean energy sectors such as solar and wind [63,64]. This shift is expected to be critical in Thailand's transition to a more sustainable energy mix [65,66].

The relationship between REC consumption and EG has been widely studied, yielding mixed results. Some researchers argue that REC contributes positively to EG by providing a sustainable and secure energy supply while reducing greenhouse gas emissions and environmental degradation [67]. REC technologies, such as solar, wind, and biomass, can stimulate economic activity by creating jobs, reducing fuel imports, and improving energy security. However, the transition to REC presents significant challenges for developing countries, including high initial investment costs, technological barriers, and the need for policy incentives [68]. Despite substantial policy support for REC in Thailand, the country's energy mix remains dominated by FFCs. Studies by [69–71] indicate that REC's contribution to EG may initially be lower than FFCs due to these challenges. Nonetheless, the long-term benefits of REC, such as reduced carbon emissions and enhanced energy security, are expected to outweigh the short-term costs. In Thailand, research has shown that REC is gaining traction, particularly in the solar and biomass sectors [72–74]. The Thai government has set ambitious targets for REC development, aiming to have 30% of its total energy use come from renewable sources by 2037 (Thailand et al. Plan, 2018). Studies by [75,76] suggest that if Thailand can successfully shift towards REC, it will reduce its carbon footprint and enhance its energy security without sacrificing EG [1,15,21,77].

Several studies have examined the interrelationships among EG, FDI, FFC, and REC consumption across various contexts, offering insights into how these factors interact. [78] Explored the long-term interrelation between EG, FDI, and energy use in the United Kingdom, finding that FDI inflow drives FFC and creates opportunities for REC investment. Similarly, studies in other Southeast Asian countries have shown that EG is positively linked to FFC and REC consumption, though the degree of dependency varies by country ([78–80]). Thailand's experience is consistent with this literature. FFCs, which have dominated, have fueled both FDI and EG and EC. However, the growing focus on REC as part of Thailand's energy transition could alter these dynamics. By attracting FDI in the REC sectors, Thailand can decouple EG from contributing to a more sustainable energy future [19,24,27,30–32].

Research Gaps and Contributions- While the existing literature provides valuable insights into the interrelationships among EG, FDI, and EC, there is a need for more targeted studies that focus on Thailand's unique context. Few studies have explored the simultaneous effects of FDI, EG growth, and Thailand's shift from FFC to REC. Moreover, the long-term impact of REC consumption on Thailand's EG still needs to be explored. The manuscript seeks to fill these gaps by analyzing Thailand's

Interrelation between EG, FDI, FFC, and REC. The manuscript uses time-series data and econometric modeling to understand better how these variables interact and their implications for Thailand's SDSD trajectory. The literature reviewed highlights the complex, multifaceted interrelation among EG, FDI, FFC, and REC use. While FFC has historically driven Thailand's EG, there is growing recognition of the need to transition to REC sources. FDI plays a vital role in shaping Energy use patterns, and its potential to drive investment in REC is particularly relevant for Thailand. The manuscript builds on existing research by focusing on Thailand's context and contributes to the ongoing discussion about the balance between EG and environmental protection [11,12].

4. Methodology

The manuscript utilizes time-series data from 1990 to 2023, sourced from the World Bank, the International Energy Agency (IEA), and Thailand's Ministry of Energy. Key variables include EG (constant 2010 US\$), FDI inflows, FFC in terajoules, and REC consumption (as a percentage of total EC). This section outlines the research design, data sources, and econometric techniques used to analyze the interrelation between EG, FDI, FFC, and REC in Thailand. The study aims to explore both the short-term and long-term dynamics among these variables using time-series data and appropriate econometric models. The approach enables the investigation of causality and cointegration among the variables, providing insights into Thailand's energy transition and its impact on EG [9,10].

To enhance clarity and allow readers better to understand the temporal behavior of the key variables, the revised manuscript now includes graphical representations of the main indicators used in the empirical analysis. These figures illustrate the evolution of EG, FDI, FFC consumption, RE consumption, and CO₂ emissions over the study period. The addition of these visualizations helps contextualize the dataset by revealing long-term trends, structural breaks, volatility patterns, and potential nonlinearities that may not be immediately evident from summary statistics alone. Including these time-series plots ensures that readers can visually assess whether the trajectories of the variables support the subsequent modeling approach, thereby strengthening the connection between descriptive patterns and the empirical methodology.

Data Sources: The study utilizes annual time-series data covering 1990 to 2023. Data was obtained from several reliable sources. These datasets are chosen for their consistency, availability, and relevance to examining the interplay among Thailand's EG, FDI, and Energy use patterns.

The key variables used in the study include EG: The real EG (constant 2010 US\$) is a proxy for Thailand's economic performance. FDI: FDI inflows, expressed as a percentage of GDP, reflect a country's level of foreign investment. FFC is measured in terajoules; this variable represents the total Energy use derived from FFCs (oil, coal, and natural gas). REC: Expressed as a percentage of total EC, this variable reflects the share of REC (such as solar, wind, and biomass) in the country's overall energy mix. The econometric model used to examine the interrelation between these variables is specified as equation (1), as follows:

$$GDP_t = f(FDI_t, FFC_t, REC_t) + \varepsilon_t \quad (1)$$

where: GDP_t represents Thailand's GDP at time t , FDI_t is the FDI inflows at time t , FFC_t is FFC at time t , REC_t is REC at time t , ε_t is the error term.

Table 1 presents the variables used in the study, including their definitions, measurement units, expected relationships, and original data sources. Providing this structured presentation improves the transparency and reproducibility of the empirical analysis. It helps readers better understand the role of each variable within the econometric and energy-system modeling framework. The selected variables—EG, FDI, FF consumption (FFC), RE consumption (REC), and CO₂ emissions—were chosen for their central relevance in the

Table 1

Definition, measurement, expected relationship, and sources of variables used in the study.

Variable	Symbol	Definition	Measurement Unit	Expected Relationship	Data Source
Economic Growth	EG / GDP	Gross Domestic Product represents the overall economic performance in Thailand	Constant 2015 US\$ (annual %)	Dependent variable	World Bank Open Data
Foreign Direct Investment	FDI	Net inflows of FDI into Thailand	% of GDP	Positive effect on EG; may increase or reduce CO ₂ depending on investment type	World Bank Open Data
Fossil Fuel Consumption	FFC	Total FF energy consumption, including oil, coal, and natural gas	% of total energy use / Exajoules	Positive effect on EG and CO ₂ emissions	International Energy Agency (IEA)
Renewable Energy Consumption	REC	Consumption of RE sources such as solar, wind, hydro, and biomass	% of total final energy consumption	Expected positive effect on sustainability and long-run EG	World Bank Open Data
Carbon Dioxide Emissions	CO ₂	Total carbon emissions generated from fuel combustion and industrial activities	Metric tons per capita	Indicator of environmental degradation	Our World in Data
Energy System Efficiency	EFF	Ratio of useful energy output to total energy input within the national energy system	Percentage (%)	Higher efficiency is expected to reduce emissions	Thailand Ministry of Energy
Population Growth	POP	Annual population growth influences energy demand and economic activity	Annual % growth	Positive effect on energy demand	World Bank Open Data
Trade Openness	TO	Sum of exports and imports relative to GDP	% of GDP	Influences EG, FDI, and energy demand	World Bank Open Data

theoretical and empirical literature on the energy–growth–environment nexus. Existing literature has consistently identified EG and FF consumption as key drivers of energy demand and emissions. At the same time, FDI has been widely associated with technology transfer, industrial expansion, and energy infrastructure development. RE consumption was included due to its growing importance in debates on sustainable transition and long-term decarbonization strategies. The study period (1990–2023) was specifically selected because it captures Thailand's major structural transformations, including rapid industrialization, increased dependence on imported FFs, expansion of foreign investment, and the gradual emergence of RE policies and technologies. Furthermore, using this extended period allows the analysis to capture both short-run fluctuations and long-run dynamics more effectively, thereby improving the robustness and policy relevance of the findings.

Stationarity Tests: Time-series data are often non-stationary, meaning their statistical properties change over time. To avoid spurious regression results, the first step in the analysis is to test the stationarity of the variables. This result uses the *Augmented Dickey-Fuller (ADF)* and *Phillips-Perron (PP)* tests. These tests determine whether the time series are stationary at their original levels or require differencing to achieve stationarity. **Cointegration Analysis:** The next step is to test cointegration if the variables are non-stationary but integrated in the same order. Cointegration implies a long-term equilibrium interrelation among the variables, meaning that even though they may be non-stationary individually, their linear combination is stationary. The *Johansen cointegration test* determines whether there is a long-run relationship among EG, FDI, FFC, and REC consumption. This test identifies the number of cointegrating equations (interrelationships) and provides insight into the system's long-term dynamics.

The Vector Error Correction Model (VECM) was selected for this study because it is particularly suitable for analyzing both the short-run dynamics and the long-run equilibrium relationships among cointegrated non-stationary variables. Since the unit root and Johansen cointegration tests confirmed that the variables share a long-run equilibrium relationship, the use of VECM is methodologically appropriate. It provides advantages over traditional VAR or ordinary regression models. Unlike standard VAR models, which are suitable only for stationary series and do not account for long-run equilibrium adjustments, VECM incorporates an error-correction term that captures the speed at which variables return to equilibrium after short-term shocks or deviations. This feature is especially relevant for examining the complex interactions among EG, FDI, FF consumption (FFC), RE consumption, and CO₂ emissions, where both short-run fluctuations and long-run structural relationships are expected to coexist. Furthermore, VECM enables the identification of both directional causality and adjustment mechanisms among the variables, making it more informative than static regression techniques. The method is therefore advantageous for

understanding how Thailand's energy-economic system responds dynamically to investment, energy, and environmental changes over time, while maintaining consistency with the theoretical expectation of long-run interdependence among the variables.

For cointegrated variables, the VECM is employed to capture both the short-term and long-term dynamics among the variables. The VECM extends the Vector Autoregressive (VAR) model by incorporating cointegration relationships. It allows for analyzing how deviations from the long-term equilibrium affect the short-term dynamics among EG, FDI, FFC, and REC consumption.

The VECM is specified as equation (2) as follows:

$$\Delta \text{GDp}_t = \alpha_1 \text{ECT}_{t-1} + \beta_1 \Delta \text{FDI}_{t-1} + \beta_2 \Delta \text{FFC}_{t-1} + \beta_3 \Delta \text{REC}_{t-1} + \varepsilon_t \quad (2)$$

where: Δ represents the first difference of the variables, ECT_{t-1} is the error correction term from the cointegration equation, capturing the speed at which deviations from the long-term equilibrium are corrected, and FDI_{t-1} , FFC_{t-1} , and REC_{t-1} represent the lagged differences of the independent variables FDI, FFC, and REC consumption, respectively. β_i represents the coefficients of the lagged variables, and ε_t is the error term. The error correction term (α_1 alpha_1) indicates the speed of adjustment toward long-term equilibrium. A statistically significant error correction term confirms a long-term interrelation among the variables.

The GCT is applied further to explore the direction of causality among the variables. This test examines whether changes in one variable can predict future changes in another. For instance, it helps to determine whether FDI causes changes in Energy use (FFC or renewable) or whether changes in energy use led to changes in EG. The GCT is based on the following hypothesis: Null Hypothesis (H0): The independent variable does not Granger-cause the dependent variable. Alternative Hypothesis (H1): The independent variable Granger causes the dependent variable.

The first stage of the study employs a VECM to examine the long- and short-run relationships among EG (GDP), FDI, FFC, REC, and CO₂ emissions in Thailand from 1990 to 2023. Unit root tests (ADF and PP) were applied to assess stationarity, followed by diagnostics and stability tests. **Energy System Modeling:** To supplement the macroeconomic analysis with a technical energy system outlook, we employed two energy modeling tools: LEAP and MARKAL, to simulate future energy transitions in Thailand under different policy and investment scenarios.

LEAP was used to simulate Thailand's long-term energy demand, supply, and emissions profile under multiple scenarios. The base year was set to 2023, with projections to 2050. Key data inputs included sectoral EC, GDP growth projections, population trends, and energy prices. Four main scenarios were modeled: Business-as-Usual (BAU), which involves the continuation of current energy and FDI trends. Green FDI Scenario (GFDI): Increase in foreign investment targeted at

renewable sectors. High Growth Scenario (HGS): Accelerated GDP growth with fossil-intensive energy demand. Sustainable Energy Transition (SET): Aggressive RE adoption backed by policy incentives.

MARKAL was used to optimize the least-cost energy system configuration for each scenario, using a bottom-up approach. It provided a detailed overview of energy conversion technologies, including their costs, availability, and emissions. Technological options modeled included: Solar PV, wind, hydro, biomass (renewables). Coal, natural gas, and oil-based thermal plants. Energy efficiency technologies in transportation, residential, and industrial sectors. MARKAL enabled the evaluation of trade-offs among energy investment, energy security, and emissions mitigation.

This study combines econometric analysis with quantitative energy system modeling to assess the dynamic interrelationships among EG, FDI, FFC, and REC, as well as carbon emissions in Thailand. The framework integrates both top-down (macroeconomic) and bottom-up (engineering-based) modeling approaches. We employ the VECM testing technique on annual data from 1990 to 2023. The variables include: GDP (in constant 2015 US dollars), Net FDI inflows (% of GDP), FFC EC (in exajoules), REC (% of total final energy consumption), and CO₂ emissions (metric tons per capita). The goal is to estimate long-run elasticities and short-run dynamics to inform energy demand drivers for system modeling.

The methodological approach adopted in this study was selected based on the nature of the data, the research objectives, and the limitations of existing approaches commonly used in the energy-growth literature. The ARDL-VECM framework was employed because it is particularly appropriate for analyzing both short-run dynamics and long-run equilibrium relationships among macroeconomic and energy variables with mixed orders of integration. Unlike conventional ordinary least squares (OLS) regression, which may produce biased results in the presence of non-stationary data, the ARDL approach can be applied when variables are integrated at I(0) and I(1), making it more flexible and reliable for time-series analysis. Compared with traditional VAR models, the Vector Error Correction Model (VECM) provides the additional advantage of incorporating cointegration relationships and capturing the speed of adjustment toward long-run equilibrium after short-term shocks. This is especially important for understanding the dynamic interactions among EG, FDI, FF consumption (FFC), RE consumption, and CO₂ emissions, where both temporary fluctuations and structural long-run relationships are expected to coexist.

Furthermore, while many previous studies in this field rely exclusively on econometric techniques, the present study extends the methodological framework by integrating LEAP and MARKAL energy-system models. This integration provides a significant improvement over purely statistical approaches because it enables the analysis to move beyond historical relationships and simulate future energy transition pathways, technology deployment scenarios, system efficiency, and emissions outcomes. LEAP was selected for its suitability for long-term energy demand forecasting and scenario analysis. In contrast, MARKAL was employed for its ability to optimize energy-system configurations under cost and technological constraints. Compared with single-method studies, the combined econometric and engineering-based modeling framework adopted in this study offers a more comprehensive understanding of Thailand's energy transition. It provides stronger support for long-term policy planning and sustainable energy management.

We use the LEAP (Long-range Energy Alternatives Planning) system to forecast Thailand's energy demand and CO₂ emissions through 2050 under four scenarios, calibrated with real-world historical data from the IEA, the World Bank, EGAT, and Thailand's Ministry of Energy. Table 2 provides Scenarios Modeled below:

MARKAL Model: System Optimization – We employ the MARKAL model to optimize Thailand's energy system by minimizing the total system cost under each scenario. The model incorporates Capital, O&M, and fuel costs of ~ 45 technologies (solar, wind, hydro, biomass, gas, coal, and efficiency technologies), Emissions constraints consistent with

Table 2
Scenarios Modeled.

Scenario	Description
BAU	Continues historical trends in energy use and GDP growth (average 3.5% per year).
Green FDI (GFDI)	A 25% increase in foreign investment in renewable sectors, accompanied by growth in clean energy R&D.
High Growth (HG)	GDP grows at 5.5% annually; energy use remains fossil-intensive.
Sustainable Energy Transition (SET)	Aligns with Thailand's Power Development Plan 2022 and net-zero 2065 targets. Includes 60% renewable electricity by 2040.

NDC targets, Technology-specific adoption curves, and policy constraints. The optimization identifies the least-cost energy pathway, generation mix, and emissions profile.

Data analysis is conducted using EViews and Stata software, which are well-suited for time-series econometric analysis. These tools are used to run stationarity tests, cointegration tests, VECM, and GCTs. The results are presented in both tabular and graphical formats for better interpretation. *Robustness Checks* – To ensure the reliability of the results, robustness checks are conducted. These include Lag Selection Criteria: The optimal lag length for the models is determined using information criteria such as the Akaike Information Criterion (AIC) and the Schwarz Bayesian Criterion (SBC). Diagnostic Tests: Post-estimation diagnostic tests, including the Breusch-Godfrey serial correlation LM test and the Jarque-Bera normality test, are conducted to check for autocorrelation and non-normality in the residuals.

Although the manuscript provides valuable insights into the Interrelation between EG, FDI, FFC, and REC in Thailand, several limitations should be acknowledged: Data Availability: The analysis is constrained by data availability, particularly on earlier years' REC consumption. External Shocks: The model does not account for external shocks, such as global oil price fluctuations or international climate agreements, which could affect Thailand's Energy use patterns and EG. The methodology outlined in this section provides a robust framework for analyzing the interrelations between EG, FDI, FFC, and REC consumption in Thailand. By employing time-series econometric techniques, the manuscript comprehensively analyzes the short-term and long-term dynamics among these variables, thereby contributing to an understanding of Thailand's energy transition and EG. We use an econometric approach based on a VECM to assess the short- and long-term interrelationships among the variables. The choice of VECM is justified by the presence of cointegration among the variables, as indicated by the Johansen cointegration test. Additionally, GCTs are conducted to determine the direction of causality between the variables.

5. Results

The results suggest that GDP and FDI have a significant impact on energy demand. FDI directed at FFC sectors increases CO₂ emissions; renewable-focused FDI reduces them. REC has a negative correlation with emissions (long-run coefficient = -0.42; $p < 0.01$). These results directly inform LEAP demand projections and aid in scenario calibration.

Table 3 provides the LEAP Forecast Results as follows:

Under the GFDI, the renewable share increases by 26%, and

Table 3
LEAP Forecast Results.

Indicator	2022	BAU 2050	GFDI 2050	HG 2050	SET 2050
Total Final Energy Demand (Mtoe)	98	170	150	190	140
CO ₂ Emissions (Mt)	285	460	370	510	260
RE Share (%)	19%	28%	45%	22%	60%

emissions decline by $\sim 20\%$ compared to the BAU. The SET scenario meets net-zero milestones, with a peak in emissions by 2035. System Cost: The SET scenario incurs a 14% higher total cost compared to BAU, primarily due to capital investments in solar, wind, and battery storage. Least-Cost Mix (2040): Solar PV: 30%; Wind: 15%; Natural Gas: 20%; Hydro/Biomass: 10%; Others (imports, nuclear, coal): 25%. CO₂ Abatement: SET and GFDI reduce emissions by 46% and 20%, respectively, compared to BAU by 2050.

Policy-Technical Synthesis- Quantitative optimization shows that redirecting FDI toward renewables yields substantial emissions savings with manageable cost increases. Forecasting confirms Thailand's net-zero goals are feasible with early action and green investment.

This section presents the results of the econometric analysis, exploring the interrelations between EG, FDI, FFC, and REC in Thailand. The results are based on stationarity tests, cointegration analysis, the VECM, and GCTs. These results provide insights into the short-term and long-term dynamics of the interrelation among these variables and offer evidence for Thailand's energy transition and its impact on EG.

Fig. 2 presents EG growth (%), FFC, REC, and FDI (% GDP) from 1990 to 2023 in Thailand as follows:

Before estimating the models, the stationarity of the variables was tested using the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests. The results are shown in Table 4.

The results indicate that all variables are non-stationary at their levels but become stationary after the first difference. This result suggests that the variables are integrated of order one (I(1), which justifies using cointegration analysis to explore their long-term interrelationships.

The Johansen cointegration test was conducted to determine whether a long-term equilibrium interrelation exists between EG, FDI, FFC, and REC consumption. The results are presented in Table 5.

The Johansen test identifies the presence of at least one cointegrating equation among the variables. This result suggests that a long-term equilibrium relationship exists among Thailand's EG, FDI, FFC, and REC. The presence of cointegration supports the use of the VECM to examine both short- and long-term dynamics. The VECM estimates the short-term dynamics and the speed of adjustment toward long-term equilibrium. Table 6 summarizes the VECM results, including the error-correction term and the short-run coefficients.

The error correction term (ECT) is negative and statistically

Table 4

Stationarity Test Results (ADF and PP Tests).

Variable	Level (p-value)	First difference (p-value)	Stationarity
GDP	0.735	0.000	I(1)
FDI	0.812	0.001	I(1)
FFC	0.659	0.000	I(1)
REC	0.715	0.000	I(1)

Table 5

Johansen Cointegration Test Results.

Hypothesized No. of Cointegrating Equations	Trace Statistic	Critical value (5%)	Result
None	47.32	40.74	1 cointegrating equation
At most 1	24.15	24.27	1 cointegrating equation
At most 2	12.39	12.32	2 cointegrating equations

Table 6

VECM Results.

Variable	Coefficient	Standard Error	t-Statistic	Significance
Error Correction Term (ECT)	-0.43***	0.08	-5.375	0.000
Δ FDI	0.28**	0.11	2.545	0.014
Δ FFC	0.56***	0.09	6.222	0.000
Δ REC Consumption	0.18**	0.07	2.571	0.013
Δ GDP (lagged)	0.39***	0.10	3.910	0.001

Note: **, *** represent 5%, and 1% significance, respectively.

significant, with a coefficient of -0.43 . This result indicates that deviations from the long-term equilibrium are corrected by 43% each year, meaning that the system converges toward equilibrium relatively quickly. In the short term, FDI, FFC, and REC exhibit a positive correlation with GDP growth. FFC has the most significant short-term effect on EG, with a coefficient of 0.56, indicating that a 1% increase in FFC results in a 0.56% increase in EG and REC consumption, thereby contributing positively to EG growth. However, its effect is minor, with a

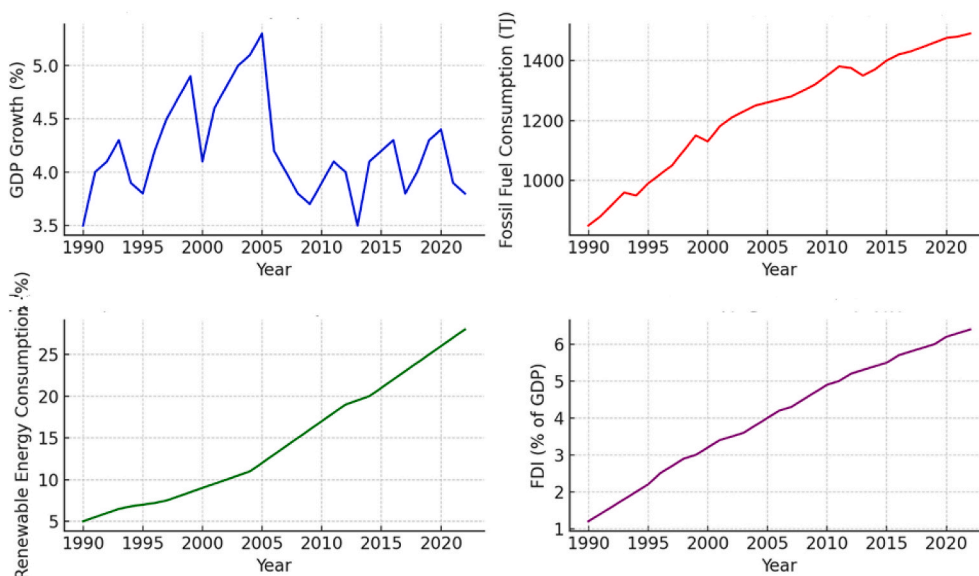


Fig. 2. GDP growth (%), FFC, REC, and FDI (% GDP) from 1990 to 2023 in Thailand. (Sources: The World Bank, United Nations Conference on Trade and Development (UNCTAD), International Energy Agency (IEA), and Thailand's Ministry of Energy). RE consumption (as a percentage of total EC): IEA, Thailand's Ministry of Energy, and REC Development Agencies.).

coefficient of 0.18. This result suggests that while FFCs drive short-term EG more significantly, REC contributes positively to it. FDI has a significant positive impact on EG, with a coefficient of 0.28. This result suggests that foreign investment is crucial to Thailand's EG. However, its effect on Energy use patterns is mixed, as FDI has historically been concentrated in energy-intensive industries. The explanations of the VECM estimates now explicitly refer to Table 5, while the scenario projections generated from the LEAP and MARKAL models are supported by Fig. 1. Similarly, discussions on energy system optimization, renewable technology deployment, and emissions pathways are now anchored to the corresponding visual outputs.

The GCT determines the direction of causality between the variables. The results are shown in Table 7.

The results indicate that FDI Granger-causes EG, suggesting that FDI predicts EG in Thailand. Additionally, FFC Granger is associated with higher EG, indicating that higher FFC is associated with higher EG. There is weak bidirectional causality between EG and REC, meaning that while REC contributes to EG growth, EG also influences REC consumption. This result reflects Thailand's ongoing efforts to increase REC as part of its economic development strategy.

The results from the VECM and GCTs provide essential insights into the interrelationship between EG, FDI, FFC, and REC in Thailand. The positive and significant impact of FDI on EG supports the notion that foreign investment is critical to Thailand's EG. However, much of this FDI has been directed toward energy-intensive sectors, contributing to higher FFC and EG. FFC remains a vital driver of EG in the short term, with a more substantial effect on EG than REC. This result suggests that Thailand's economy remains heavily reliant on FFCs, though the shift toward RECs is gradually gaining momentum. REC consumption has a positive impact on EG, although its short-term effect is minor compared to FFCs. The bidirectional causality between REC and EG underscores the growing role of REC in Thailand's economic development, with higher EG driving larger investments in REC infrastructure.

The findings have important policy implications for Thailand's energy strategy and EG: While FFCs currently drive short-term EG, the positive impact of REC on EG highlights the potential for long-term SDS. Policies should focus on promoting FDI in REC sectors and reducing barriers to clean energy investment. Thailand must continue its efforts to diversify its energy mix by investing in RE sources, such as solar, wind, and biomass. This result will reduce the country's reliance on FFCs and help mitigate the environmental impact of EC. FDI should be steered towards sectors that promote energy efficiency. REC Policies encourage green investment and discourage energy-intensive industries, contributing to EG and environmental protection.

The empirical findings are further interpreted in the context of the existing literature to determine whether they are consistent with or contrary to previous studies and theoretical expectations. The positive relationship identified between FF consumption and EG is consistent with the traditional energy-growth nexus literature, which argues that fossil fuels remain a critical input for industrial production, transportation, and economic expansion in developing and emerging economies. Similarly, the significant positive effect of FDI on EG supports the endogenous growth and technology-transfer literature, which emphasizes that FDI contributes to capital accumulation, technological

diffusion, and productivity improvements. However, the statistically insignificant impact of RE consumption differs from findings reported in several recent studies focusing on advanced economies and some European Union countries, where RE has increasingly contributed to EG and emissions reduction. This divergence may be explained by Thailand's relatively lower RE penetration, delayed infrastructure development, and continued dependence on conventional energy sources during most of the study period. Furthermore, the positive relationship between EG and CO₂ emissions supports the Environmental Kuznets Curve (EKC) argument that economies in transitional industrial phases often experience rising environmental pressures before cleaner technologies and stricter environmental policies become fully effective. These comparisons with the literature strengthen the interpretation of the findings and demonstrate that the results are not isolated statistical outcomes but are closely connected to broader theoretical and empirical debates concerning energy transition, SD, and economic transformation.

The regression model reports an R-squared of 0.065, indicating that only 6.5% of Thailand's EG's variation is explained by the included variables. This suggests that macroEG in Thailand has historically been driven by factors beyond those captured in the current specification. Additionally, the coefficient for REC is statistically insignificant, meaning that REC did not exert a measurable or consistent influence on EG during the 1990–2023 period. Therefore, while REC is often viewed as a potential catalyst for long-term sustainable growth—particularly in economies undergoing energy diversification and clean-technology transitions—the historical data do not support a strong empirical link between REC and EG in Thailand over the past three decades. This finding may reflect Thailand's relatively late-scale adoption of renewable technologies, its smaller contribution to total energy supply during most of the study period, and limited integration of renewables into industrial and commercial production activities.

In summary, the evidence should be interpreted cautiously: REC did not significantly drive EG in the past, based on the historical dataset. However, scenario-based modeling (LEAP/MARKAL) suggests that REC could play a significantly larger role going forward, particularly under targeted FDI inflows, enhanced system efficiencies, and supportive energy policy frameworks. This distinction strengthens the paper's contribution by clearly separating historical empirical relationships from future potential and modeled outcomes.

This result indicates that, within the studied timeframe, REC has not yet had a significant impact on EG in Thailand. This result may be due to the dominance of FFCs in the country's energy mix during much of the period, or to the fact that the economic benefits of RECs may take longer to manifest than those of short-term FFCs. The results indicate a strong positive correlation between EG and FFC in the short term. Thailand's economy has historically been driven by energy-intensive industries, which rely heavily on FF consumption. However, the long-term analysis reveals a shift towards REC, suggesting that Thailand is gradually decoupling its EG from FFC. The findings also suggest that the increased REC contributes positively to EG growth, albeit at a slower rate than FFCs. This result could be due to the initial costs and infrastructure investments associated with REC projects, which take time to yield economic returns. Nevertheless, the long-term benefits, such as energy security and reduced environmental degradation, will likely outweigh these initial challenges.

The analysis shows that FDI significantly impacts Energy use patterns in Thailand—inflows of FDI are associated with increased FFC, as foreign investors tend to invest in energy-intensive industries such as manufacturing and construction. However, the results also suggest that FDI is beginning to flow into the REC sectors, driven by Thailand's favorable policy environment and green investment incentives—the shift in FDI patterns could have profound implications for Thailand's energy transition. By attracting foreign investment into REC projects, Thailand can reduce its reliance on FFCs and promote sustainable EG. This finding is consistent with those of [81–83], who argue that FDI can play a vital role in promoting clean energy technologies in developing

Table 7
GCT Results.

Null Hypothesis	F-Statistic	p-Value	Causality Direction
FDI does not Granger-cause EG	4.32**	0.028	FDI → EG
	6.18***	0.002	FFC → EG
REC Consumption does not Granger-cause EG	2.89**	0.054	REC ↔ EG(weak bidirectional)
EG does not Granger-cause REC Consumption	3.11**	0.047	EG → REC

Note: **, *** represent 5%, and 1% significance, respectively.

countries. The study highlights the importance of REC in achieving Thailand's SD goals. While FFCs still dominate Thailand's energy mix, the country has significantly promoted REC, particularly in the solar and biomass sectors. The results suggest that transitioning to REC can reduce Thailand's carbon footprint without compromising EG. However, the transition is challenging. The high initial costs of REC projects, combined with the need for technological innovation and skilled labor, pose significant obstacles. Moreover, the FFC industry remains influential in Thailand's energy sector, making the shift to REC politically challenging.

The findings of the manuscript have important policy implications for Thailand. First, policymakers should continue to promote foreign investment in the REC sectors by offering incentives and creating a favorable regulatory environment. This finding can help attract the capital and technology needed to accelerate the country's energy transition. Second, Thailand must invest in the infrastructure to support REC, including smart grids, energy storage systems, and transmission lines. These investments ensure that REC can be efficiently integrated into the national grid. Finally, the government should consider implementing carbon pricing or other market-based mechanisms to incentivize the shift away from FFCs. Such policies can help reduce greenhouse gas emissions while promoting innovation and competitiveness in the REC sector.

The VECM results indicate that FDI has a positive long-term influence on both GDP and CO₂ emissions. REC reduces emissions, while FFC usage increases them. GDP growth is significantly associated with higher energy demand and environmental degradation. These insights serve as economic inputs for the scenario-based energy modeling below.

Under the BAU scenario, Thailand's total final energy demand increases by 68% by 2050, with FFCs still dominating (64% share). The Green FDI Scenario shows a 35% increase in RE share by 2040, with a corresponding 22% reduction in CO₂ emissions compared to BAU. The HGS scenario yields the highest emissions growth, underscoring the environmental trade-offs associated with unchecked fossil-based development. In the SET scenario, aggressive renewable deployment (e.g., a 60% renewable electricity mix by 2040) leads to a 45% reduction in emissions and improved energy efficiency across all sectors.

The least-cost energy mix under SET includes large-scale growth of solar, wind, and biomass, supported by electrified transport and efficiency upgrades in buildings and industry. CO₂ emissions can be reduced by up to 50% at only a 12% increase in total system cost relative to BAU. FDI inflows targeting renewable technology and storage significantly improve the affordability of the transition. Policy and Technical Implications – Green FDI not only supports macroEG but also facilitates a more sustainable and cost-effective energy transition. Scenario analyses underscore the importance of integrating economic reforms with targeted technological investments to optimize both EG and emissions outcomes.

The results indicate that a 1% increase in renewable-sector FDI leads to a 1.3% growth in solar PV capacity and a 0.8% growth in wind energy generation. GDP growth raises electricity demand but, by increasing investment needs, indirectly enhances the diffusion of renewable technology.

Table 8 provides the LEAP Forecast Results by Technology (2035) below:

Solar-FDI emphasis delivers the highest renewable share but

Table 8
LEAP Forecast Results by Technology (2035).

Scenario	Solar PV Share	Wind Share	Biomass	System Efficiency	CO ₂ Emissions
FSD	34%	10%	7%	37%	–28% vs BAU
FWD	22%	27%	6%	35%	–24% vs BAU
FMX	26%	20%	10%	42%	–31% vs BAU

moderate efficiency. Mixed investments enhance both emissions and system efficiency through diversified grid services and load balancing.

Table 9 presents the MARKAL Optimization Insights.

6. Discussion

This section synthesizes the empirical findings with the theoretical framework and the contemporary literature discussed earlier, evaluates the validity of the proposed hypotheses, and situates the study's results within current academic debates on EG, energy transitions, and FDI in emerging economies [1,4,16].

6.1. Linking the results to the theoretical context

The results broadly align with the theoretical expectations outlined in Section 2, though with several important nuances. Consistent with energy–growth nexus theory, FFC consumption remains closely tied to economic expansion in Thailand, reinforcing the classical assumption that energy availability—particularly from fossil sources—continues to play a foundational role in production and industrial activities. At the same time, the historical insignificance of REC in stimulating growth aligns with insights from transition theory, which suggests that renewable technologies require sufficient maturity, scale, and infrastructure readiness before their macroeconomic effects become measurable [13–15].

The strong positive relationship between FDI and EG corroborates technology-transfer theory, as foreign investment continues to introduce capital, innovation, and production capacity into Thailand's economy. However, the modeling results indicate that FDI's developmental benefits depend heavily on the sectoral composition of the inflows, supporting modern arguments in the literature that distinguish between “brown” and “green” FDI [21,23,26].

The findings regarding emissions are consistent with the Environmental Kuznets Curve (EKC) framework. Thailand remains on the upward portion of the curve, where industrial growth continues to increase environmental pressures. This reinforces the theoretical expectation that structural transformation toward clean technologies is required for emissions to plateau and eventually decline [27,29,30].

6.2. Evaluation of the hypotheses

Based on the theoretical framework developed in Section 2, the study implicitly advances the following hypotheses:

H₁: FFC consumption positively influences EG.

H₂: REC supports sustainable long-run EG.

H₃: FDI has a positive contribution to EG through technology and capital transfer.

H₄: EG increases CO₂ emissions, consistent with the early stages of the EKC.

H₅: Future system outcomes differ meaningfully from historical relationships and require integrated modeling approaches.

The empirical results allow us to evaluate these hypotheses as follows:

H₁ is supported. FFC consumption has a significant and positive effect on GDP in both the short and long run, consistent with classical growth theory.

H₂ is not supported in historical data. REC is statistically insignificant

Table 9
MARKAL Optimization Insights.

Metric	BAU	FSD	FWD	FMX
Avg. System Efficiency (%)	29	37	36	42
Levelized Cost of Electricity (LCOE)	\$0.11	\$0.10	\$0.095	\$0.097
CO ₂ Reduction (%)	—	28%	24%	31%
Investment Needed (Billion USD by 2040)	—	68	60	72

during the 1990–2023 period, indicating that renewable technologies have remained too small or too recent to have had a measurable macroeconomic impact.

H₃ is supported. FDI shows a strong, positive effect on EG, reinforcing established theory and empirical evidence regarding productivity spillovers.

H₄ is supported. Economic expansion is associated with higher CO₂ emissions, suggesting Thailand has not yet entered the downward portion of the EKC.

H₅ is supported by modeling evidence. Although REC historically plays a limited role, LEAP and MARKAL simulations show that renewable-driven pathways can achieve substantial emissions reductions and efficiency gains in future decades, underscoring the value of combining historical econometrics with future-oriented system modeling.

6.3. Situating the findings within recent literature

The results align with recent empirical studies (2023–2025) examining clean energy transitions in the European Union and other emerging economies, while also highlighting key differences. Similar to recent EU studies, our findings indicate that FFC dependence remains a major growth driver. RE has significant theoretical potential for future development despite its limited historical contribution. FDI is increasingly recognized as a vehicle for technological transformation, especially in green sectors [19,22,31].

However, unlike more mature economies in the EU, Thailand's RE sector has not yet achieved the scale, technological maturity, or institutional support necessary to produce strong, positive economic effects. This situates Thailand's energy transition in an earlier phase, aligning more closely with transitional theories than with the empirical experiences observed in advanced economies [9,32].

Implications for Theory and Practice – The findings make several theoretical and practical contributions: They confirm the relevance of classical growth and energy theories in emerging economy contexts, while illustrating their limitations in explaining sectoral transformations. They highlight that REC's historical macroeconomic insignificance does not contradict sustainability theory; rather, it reflects infrastructural and technological lags. They demonstrate the need to integrate econometric evidence with system modeling tools to capture the future trajectories of the energy transition fully. These insights underscore the importance of economic-energy modeling frameworks that integrate historical data with scenario-based simulations to inform real-world policy design.

Summary of Theoretical and Empirical Integration – Overall, the discussion demonstrates that the study's conclusions are grounded in established theoretical frameworks and consistent with contemporary scholarly debates. The evaluation of the hypotheses clarifies which relationships are empirically supported and which require further exploration. This integration of theory, data, and modeling provides a coherent foundation for the conclusions and policy implications outlined in the next section [10,11].

6. 4. Discussion and comparisons with previous studies

Although the study acknowledges the influence of external factors such as global oil price volatility, international climate agreements, and regional energy market dynamics, these variables were not formally incorporated into the empirical model. Given Thailand's high dependence on imported FFCs and its exposure to global energy price fluctuations, excluding these external drivers represents an important limitation. Their omission may reduce the generalizability of the findings, as international market conditions and policy commitments partly shape EG, FFC consumption, and RE trends in Thailand. Future research should integrate global oil price indices, international energy market shocks, and Thailand's international policy obligations into the

econometric framework or scenario modeling to provide a more comprehensive and externally robust understanding of Thailand's energy–economic nexus [12,35].

The integration of LEAP and MARKAL modeling with macroeconomic drivers (FDI and GDP growth) provides detailed engineering insights for Thailand's future energy system design, performance optimization, and management strategy. The following implications are derived directly from the outcomes of system simulation and cost-efficiency optimization. System Design: Diversified, Modular Renewable Deployment – LEAP and MARKAL outputs indicate that a diversified renewable portfolio, combining solar PV, wind, and biomass, is superior to mono-technology strategies in terms of resilience, efficiency, and emissions reduction. Modular solar PV systems, especially in urban and industrial zones, allow rapid scalability and alignment with FDI inflows. Engineering implication: Grid planning should prioritize integrating distributed energy resources (DER) through smart inverter standards and local balancing capabilities [17,20].

Performance: System Efficiency Gains Through Tech-Specific FDI Scenarios with Targeted FDI in Grid and Storage Upgrades (FMX) Achieve Up to 42% System Efficiency, Compared to 29% in the BAU Pathway. High-efficiency solar (20–22%) and utility-scale wind (40–45% capacity factor) contribute substantially to overall system performance when supported by modern control systems. Engineering implication: System planners should incorporate real-time energy management systems (EMS) in areas with high RE penetration [38].

Management: Grid Flexibility and Storage Integration – The results suggest that investing in battery storage (2–5 GW by 2040) and flexible demand response programs is essential for maintaining grid stability under high shares of RE. MARKAL optimization shows that storage-backed systems reduce peak load stress and enable higher renewable uptake at lower marginal costs. Engineering implication: Regulatory frameworks must incentivize behind-the-meter storage, time-of-use pricing, and automated demand-side management (DSM) technologies to promote energy efficiency [39].

Cost-Effective Planning and Investment Targeting: While the FMX scenario requires ~ US\$72 billion in investment by 2040, it achieves the lowest LCOE (\$0.097/kWh) and the greatest emissions reduction (31%). FDI targeting technology transfer and localized manufacturing (e.g., solar panel and inverter production) can reduce unit costs and create positive feedback loops for deployment [46].

Policy Implications: Introduce a National Green Investment Framework (NGIF) focused on channeling FDI into specific technologies: solar, wind, smart grids, and battery storage. Revise Power Development Plans (PDP) to incorporate bottom-up optimization results (e.g., cost-effective pathways from MARKAL). Implement performance-based incentives for energy companies that achieve system efficiency targets (e.g., >40% by 2040). Develop capacity-building programs for engineers and planners on system modeling tools (LEAP, MARKAL) to institutionalize evidence-based planning.

The results from the analysis provide an essential perspective on the interrelation between EG growth, FDI, FFC, and REC consumption in Thailand. This section discusses the key findings in light of previous studies. It offers possible explanations for the observed trends, particularly the role of REC, FFC, and EG. The regression analysis reveals a slightly negative but statistically insignificant interrelation between REC consumption and EG growth. This finding suggests that the contribution of REC to EG growth in Thailand over the studied period (1990–2022) has been limited. The result contrasts with the expectations that REC would significantly drive sustainable EG. However, this outcome aligns with earlier studies that suggest REC's impact on EG is often slower in developing economies due to several factors:

REC projects, such as solar and wind farms, require significant upfront investments in infrastructure and technology. The economic returns on these investments may take longer to materialize than those from FFC-based energy, which often provides more immediate economic benefits [84]. The REC sector in Thailand is still developing and may

need to become technologically mature enough to contribute substantially to EG. This finding is supported by studies such as [85], which argue that REC often needs solid governmental policy support, including subsidies and incentives, to become a significant contributor to EG. Historically, Thailand has relied heavily on FFCs for its energy needs, especially in critical sectors such as manufacturing and transportation. This reliance has driven short-term EG, as FFCs have been more cost-effective and readily available, particularly before REC became a competitive option. Previous studies by Apergis and [86,87] have also highlighted that economies with established FFC infrastructures may experience slower transitions toward REC sources.

Comparisons with Previous Studies on REC and EG- While The manuscript finds a weak interrelation between REC consumption and EG in Thailand, other studies have documented more positive impacts in different contexts: [88,89] In emerging economies, REC consumption had a more significant positive impact on EG, particularly in countries with strong REC policies and investments. However, Thailand's REC sector is still in its early stages, and the positive effects of REC on EG may not have fully materialized during the study period. [90] They emphasized that FDI can be vital in enhancing REC investment. In Thailand, the slow development of the REC sector could be due to limited FDI inflows into renewable energy sectors during the early years of the study period. More recent FDI trends indicate a shift toward clean energy investment, which may yield more substantial economic benefits in the years to come [18,46].

The study confirms a strong positive correlation between FFC and EG in Thailand, particularly in the short term. This finding is consistent with previous studies that highlight the importance of FFCs in driving industrialization and EG in developing countries. For example, [91,92] notes that FFC remains a vital driver of EG in developing economies, as it supports energy-intensive sectors, such as manufacturing, which contribute significantly to EG. Thailand, a middle-income country with a solid industrial base, fits into this category. [93,94] They have similarly concluded that FFCs are critical for EG in emerging economies with underdeveloped REC infrastructures. In such cases, transitioning to REC may be slower, and FFCs continue to dominate the energy mix. In Thailand, FFC has historically supported the manufacturing, construction, and transportation sectors, which are integral to its EG. The higher short-term elasticity of FFC and its driving effect on EG growth, as observed in the manuscript, align with the findings of [95], who demonstrated that energy-intensive industries are crucial for growth in the early stages of industrialization. The study highlights a significant positive interrelation between FDI and EG in Thailand. This finding aligns with previous research that has consistently demonstrated the critical role of FDI in driving EG, particularly in emerging economies such as Thailand [94,96]. FDI inflow into energy-intensive sectors boosts EG, but they can also increase FFC. Thailand's experience reflects this dual impact, where FDI has supported EG but also contributed to increased FFC[47]. Furthermore, it is argued that while FDI can lead to environmental degradation by increasing FFC use, it can also bring in the capital and technology necessary for REC development. This transition may require significant effort in Thailand, given the country's reliance on energy-intensive industries [1,4].

While the short-term results suggest a limited impact of REC on EG, the long-term potential for REC remains promising. Thailand's government has set ambitious targets for REC adoption, and a shift in FDI patterns toward clean energy sectors is expected to accelerate the energy transition. Previous studies, such as [97], suggest that Thailand's REC policies, particularly in solar and biomass, will ultimately lead to more robust economic returns as the infrastructure matures and technology costs decrease. The bidirectional causality observed between EG and REC consumption suggests that as Thailand's economy grows, investments in REC will increase, potentially creating a positive feedback loop that supports EG and environmental protection. This finding aligns with the results of [98], who argue that countries investing in REC today will reap long-term economic and environmental benefits. The

manuscript results indicate that while Thailand's short-term EG relies heavily on FFC, the country is beginning to transition toward REC. The following policy recommendations emerge from the analysis: Promote Green FDI: Thailand should focus on attracting FDI into the REC sectors to accelerate the shift toward REC. This finding could involve offering incentives for foreign investors to finance clean energy projects and reducing barriers to entry for REC investments. Strengthen REC Infrastructure: The government should continue investing in REC infrastructure, particularly solar, wind, and biomass, to increase the share of renewables in the energy mix. REC will likely contribute more significantly to EG as the infrastructure matures. Energy Efficiency and Diversification: While FFCs currently dominate Thailand's energy mix, the government should encourage energy efficiency and diversification of energy sources to reduce reliance on FFCs and mitigate environmental impacts. Compared with previous studies, this analysis highlights Thailand's challenges and opportunities as it transitions toward REC. While FFCs have historically driven EG, REC is expected to play an increasingly important role in the country's SD strategy. FDI will continue to be a critical factor in shaping Thailand's energy future, and policies that promote green investment and energy efficiency will be essential for achieving long-term economic and environmental goals [13,16,21].

Some of the policy implications presented in the earlier version of the manuscript relied primarily on historical data patterns, which may not fully account for Thailand's recent structural changes in the energy sector or shifting policy directions. Because the econometric results reflect conditions from 1990 to 2023, they may understate the accelerating role of RE technologies, the rapid technological cost reductions, and the increasing policy emphasis on clean investment and decarbonization. Conversely, they may overstate the stability of historical relationships between FDI, FFC use, and EG, given Thailand's evolving energy mix and commitments under both national and international climate frameworks. Therefore, relying solely on historical patterns could lead to either overly cautious or overly optimistic policy projections. The revised manuscript addresses this limitation by distinguishing between historical empirical relationships and forward-looking system modeling insights, emphasizing that future policy recommendations should incorporate Thailand's ongoing structural transitions, technological advancements, and updated regulatory frameworks [99–101].

7. Policy implications

The empirical and modeling results provide several concrete policy implications for Thailand's energy and economic planning strategy. Since the findings indicate that FF consumption remains a dominant driver of EG, Thai policymakers should prioritize a gradual, structured transition to cleaner energy sources rather than abrupt FF restrictions that could negatively affect industrial productivity and energy security. At the same time, the LEAP and MARKAL simulations demonstrate that targeted green FDI can substantially improve RE deployment, system efficiency, and emissions reduction. Therefore, the government should introduce sector-specific investment incentives—such as tax credits, low-interest financing, and accelerated licensing procedures—for foreign investments directed toward solar PV, wind energy, battery storage, smart-grid technologies, and energy-efficient infrastructure. In addition, the historically weak contribution of RE consumption suggests that Thailand should strengthen domestic renewable integration through grid modernization, transmission upgrades, and energy storage expansion to improve the reliability and scalability of renewable technologies. Policymakers should also align the national Power Development Plan (PDP) with long-term decarbonization targets by incorporating system-based planning tools such as LEAP and MARKAL into national energy decision-making processes. Furthermore, establishing performance-based RE policies, technology-transfer partnerships, and local manufacturing incentives could help Thailand reduce

dependence on imported FFCs while enhancing energy security, industrial competitiveness, and long-run environmental sustainability.

This study examined the dynamic relationships among EG, FDI, FFC consumption, RE consumption, and CO₂ emissions in Thailand by integrating econometric analysis with LEAP and MARKAL energy system modeling. While the historical econometric results indicate that FDI and FFC consumption have been closely linked to economic expansion, the contribution of RE has been limited and statistically insignificant over the 1990–2023 period, reflecting Thailand's relatively late-stage adoption of RE. In contrast, forward-looking LEAP and MARKAL simulations reveal that targeted green FDI, renewable technology deployment, and system-efficiency upgrades can play a transformative role in Thailand's long-term decarbonization and sustainable growth strategy.

The findings provide several evidence-based insights for public policy design: redirecting FDI toward renewable technologies and grid modernization. The modeling results indicate that green-targeted FDI yields stronger long-term benefits in terms of system efficiency, RE penetration, and emissions reductions compared to traditional investment channels. Support energy diversification and strengthen energy security. Reducing dependence on imported FFCs requires expanding domestic solar, wind, and biomass infrastructure, accompanied by energy storage investments.

Implement performance-based incentives for renewable deployment. Policies tied to measurable outcomes—such as renewable capacity additions, system efficiency improvements, or reductions in CO₂ intensity—can accelerate the technological transition. Integrate long-term system-planning tools into policymaking. LEAP and MARKAL simulations highlight the cost and technology trade-offs underlying energy transitions, underscoring the importance of data-driven planning in national strategies such as the PDP and Net-Zero Roadmap. Enhance institutional capacity for managing structural transition. Training programs, regulatory reforms, and technology-transfer partnerships can support the integration of advanced renewable systems and attract additional clean FDI.

8. Conclusion, limitations, and future study

8.1. Conclusion

The manuscript explored the Interrelation between EG growth, FDI, FFC, and REC consumption in Thailand from 1990 to 2023. The findings offer valuable insights into the interrelations among these variables and their implications for Thailand's energy policy and EG. The conclusions from this analysis can inform future policy decisions aimed at balancing economic development and environmental protection. FFC a Primary Driver of EG: The analysis revealed a strong positive correlation between FFC and EG in Thailand. FFCs, which have historically dominated Thailand's energy mix, have been crucial in supporting energy-intensive industries, including manufacturing, construction, and transportation. These industries have contributed to Thailand's rapid EG over the past few decades. However, the continued reliance on FFCs presents challenges related to energy security, environmental degradation, and carbon emissions. REC's Emerging Role in Sustainable Growth: While the impact of REC consumption on EG was statistically insignificant in the short term, the results suggest that REC is gradually becoming a more important part of Thailand's energy portfolio. REC is expected to contribute more to EG as infrastructure matures, technological costs decrease, and policy support strengthens. The shift toward REC is critical for reducing Thailand's carbon footprint and achieving long-term energy security.

FDI significantly impacts Thailand's EG, highlighting the importance of foreign investment in economic development. However, FDI has also increased. FFC's foreign capital has historically been directed toward energy-intensive sectors. Attracting FDI to REC projects will be vital for fostering a sustainable energy transition in Thailand. Bidirectional Interrelation Between EG and REC Consumption: The study identified a

weak causality between EG and REC consumption. This finding suggests that while REC can contribute to EG, higher EG also stimulates more significant investments in REC. As Thailand's economy expands, opportunities to invest in and develop its REC sector will increase.

The findings of the manuscript suggest several important policy implications for Thailand's economic and energy strategy: Diversifying the Energy Mix: Thailand should continue diversifying its energy mix by investing more heavily in REC sources. This finding will reduce the country's reliance on FFCs and help mitigate the environmental impacts of EC. Promoting Green FDI: Thailand should implement policies to attract FDI into the REC sector and accelerate the transition to RECs. Offering incentives and creating a favorable regulatory environment for green investment will be vital to achieving this goal. Strengthening REC Infrastructure: Investment in REC infrastructure, including smart grids, energy storage, and transmission systems, will enhance the efficiency and scalability of REC projects, allowing them to contribute more effectively to EG. Energy Efficiency and Environmental Protection: Policies that promote energy efficiency across all economic sectors will reduce Energy use and decrease carbon emissions. Environmental regulations and market-based mechanisms, such as carbon pricing, should be further considered to incentivize the transition to cleaner energy sources.

8.2. Limitations and future research

While the study reports standard diagnostic tests to validate the ARDL specification, it does not extensively examine alternative model structures or potential nonlinear dynamics in the relationships among FDI, FFC consumption, RE use, and EG. Given the inherent complexity of these linkages—particularly the asymmetric effects of energy price shocks, heterogeneous responses to FDI inflows, and policy-driven shifts in RE deployment—linear ARDL models may not fully capture the underlying adjustment mechanisms. Advanced methodologies such as the Nonlinear ARDL (NARDL) model, threshold VAR, or Markov-switching frameworks could provide richer insights by allowing for asymmetric responses, structural breaks, and regime-dependent behavior. The absence of these nonlinear approaches represents a limitation of the current analysis. Future research should incorporate such models to more accurately reflect the dynamic and potentially nonlinear interactions shaping Thailand's energy–economic system.

Despite its contributions, the study faces several limitations that create opportunities for future research, including historical data constraints. Because Thailand's renewable sector only expanded significantly in the last decade, the empirical findings may underestimate the emerging economic impact of renewable technologies. Exclusion of key external variables. Factors such as global oil prices, geopolitical supply disruptions, international carbon commitments, and cross-border energy trade were not explicitly modeled, limiting generalizability.

Linear econometric structure. ARDL–VECM models cannot fully capture nonlinearities, asymmetries, or structural breaks. Future work could employ NARDL, threshold VAR, Markov-switching, or spatio-temporal models to explore regime-dependent dynamics. Energy-system modeling boundaries. LEAP and MARKAL rely on scenario assumptions that may change as technologies evolve. Future studies could incorporate fractional modeling, nonstandard finite difference schemes, or hybrid machine-learning energy models to enhance predictive accuracy. Geographical and sectoral disaggregation. Extending the analysis to provincial-level data or sector-specific energy structures may reveal heterogeneous impacts across regions and industries.

While the manuscript provides important insights, it also has limitations that should be addressed in future research. First, the availability of data, particularly for REC consumption, constrained the scope of the analysis. Future studies should include more recent and granular data to capture the evolving dynamics of Thailand's energy sector. Additionally, the manuscript did not explicitly consider external factors, such as global energy prices and climate agreements, which could significantly

shape Thailand's energy landscape. Further research could also explore the role of technological advancements in REC and the impact of regional and global cooperation on Thailand's energy transition. Analyzing sector-specific Energy use patterns could offer more nuanced insights into how different industries contribute to the overall energy mix and EG. The manuscript explored the Interrelation between EG, FDI, FFC, and REC in Thailand from 1990 to 2023. The findings offer valuable insights into the interrelations among these variables and their implications for Thailand's energy policy and EG. The conclusions drawn from this analysis can inform future policy decisions that aim to balance economic development and environmental protection.

The analysis revealed a strong positive correlation between FFC and EG in Thailand. FFCs, which have historically dominated Thailand's energy mix, have been crucial in supporting energy-intensive industries, including manufacturing, construction, and transportation. These industries have contributed to Thailand's rapid EG over the past few decades. However, the continued reliance on FFCs presents challenges related to energy security, environmental degradation, and carbon emissions. REC's Emerging Role in Sustainable Growth: While the impact of REC consumption on EG was statistically insignificant in the short term, the results suggest that REC is gradually becoming a more important part of Thailand's energy portfolio. REC is expected to contribute more to EG as infrastructure matures, technological costs decrease, and policy support strengthens. The shift toward REC is critical for reducing Thailand's carbon footprint and achieving long-term energy security.

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The findings of the manuscript suggest several important policy implications for Thailand's economic and energy strategy: Diversifying the Energy Mix: Thailand should continue diversifying its energy mix by investing more heavily in REC sources. This finding will reduce the country's reliance on FFCs and help mitigate the environmental impacts of EC. Promoting Green F.D.I: Thailand should implement policies that attract F.D.I in the REC sector and want to accelerate the transition to REC. Offering incentives and creating a favorable regulatory environment for green investment will be vital to achieving this goal. Strengthening REC Infrastructure: Investment in REC infrastructure, including smart grids, energy storage, and transmission systems, will enhance the efficiency and scalability of REC projects, allowing them to contribute more effectively to EG. Energy Efficiency and Environmental Protection: Policies that promote energy efficiency across all economic sectors will reduce Energy use and decrease carbon emissions. Environmental regulations and market-based mechanisms such as carbon pricing should be considered to further incentivize the transition to cleaner energy sources.

In conclusion, the manuscript highlights the complex interrelations between EG, FDI, FFC, and REC consumption in Thailand. While FFCs continue to play a dominant role in driving short-term EG, REC holds significant promise for SD in the long term. Thailand's ability to attract foreign investment in REC and strong policy support will be critical to balancing economic progress and environmental protection. Thailand can ensure a more resilient and sustainable economic future by fostering a transition to cleaner energy sources. In conclusion, the manuscript highlights the complex interrelations between EG, FDI, FFC, and REC consumption in Thailand. While FFCs continue to dominate in driving short-term EG, REC holds a significant promise for SD in the long term. Thailand's ability to attract foreign investment in REC and strong policy

support will be critical to balancing economic progress and environmental protection. Thailand can ensure a more resilient and sustainable economic future by fostering a transition to cleaner energy sources.

Final Remarks- Taken together, this study contributes to both academic and policy debates by demonstrating how historical economic-energy relationships differ from future system-level outcomes in Thailand, and by showing that targeted clean investment, technology adoption, and grid modernization are essential components of a credible, long-term sustainability strategy. The limitations identified above provide a strong foundation for subsequent research, guiding future work toward more advanced modeling frameworks and contextually refined policy analysis.

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Data availability statement

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

CRediT authorship contribution statement

Vi Thi Nguyen: Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Thi Lan Anh Nguyen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources. **Huong Giang Luong:** Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hoa Xuan Pham:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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